Hate Speech Detection through Explainable AI and Reward-Optimized LLMs

Devansh Kumar and Ayush and Saksham Tyagi and Mudit Agrawal
Project PJ-03
Bennett University
Supervisor: Dr. Pradeepika Verma

Abstract

Hate speech detection systems are increasingly deployed to assist human moderators, yet two gaps limit their practical value: most models give no faithful account of *why* a post was flagged, and multilingual code-mixed content—notably Hinglish (Hindi–English)—remains under-served. We present a four-phase study on English and Hinglish social media text. First, we establish baselines from TF–IDF classifiers to multilingual encoders (mBERT, XLM-R, MuRIL) on HateXplain. Second, we LoRA-fine-tune a small instruction-tuned LLM to emit a structured *label + rationale* pair whose format is machine-checkable. Third, we optimise this model with Group Relative Policy Optimization (GRPO) under a composite reward $R = \alpha R_{\text{label}} + \beta R_{\text{faith}} - \gamma R_{\text{bias}}$, where R_{faith} is a masking-based faithfulness score that penalises decorative rationales and R_{bias} penalises rationales that merely cite identity terms. Finally, we audit all models with comprehensiveness/sufficiency faithfulness metrics, subgroup AUC and BPSN/BNSP unintended-bias metrics, and HateCheck functional tests. Our baseline already shows measurable identity-term bias (e.g., BPSN of 0.65 for anti-Jewish content), motivating the bias-aware reward.

1 Introduction

Online hate speech causes real harm, and human moderation does not scale. Automated classifiers help, but a bare probability score gives moderators little to act on and gives researchers little to audit. If a system says a post is hateful *because of specific words*, both the moderator and the auditor can check that claim. Meanwhile, moderation demand in India and similar multilingual markets involves *code-mixed* text—Roman-script Hindi–English (Hinglish)—where most English-only systems degrade.

This project makes three contributions:

1. A **structured rationale format** for instruction-tuned LLMs (`Label:/Rationale:`) whose parseability, label accuracy and rationale faithfulness can each be measured and rewarded (§3).

2. A **composite reward** for GRPO post-training that combines label accuracy with a masking-based *faithfulness* term and an identity-term *bias penalty*, with tunable weights α, β, γ (§3.4).

3. A **bias-aware evaluation protocol**: comprehensiveness and sufficiency for rationales; subgroup AUC, BPSN and BNSP (Borkan et al., 2019) per identity group; HateCheck functional tests (Vidgen et al., 2021); and out-of-distribution transfer (§3.5).

2 Related Work

Hate speech datasets. HateXplain (Mathew et al., 2021) provides English Twitter/Gab posts with three-annotator labels, rationale spans and target communities—the rationale spans are what make explainability trainable. HASOC (Mandl et al., 2020) covers Hindi, English and code-mixed tracks. HateCheck (Vidgen et al., 2021) provides 3,728 functional test cases across 29 functionalities, including identity-term usage in non-hateful contexts—exactly the failure mode our bias penalty targets.

Rationales and faithfulness. ERASER (DeYoung et al., 2020) established comprehensiveness and sufficiency as the standard faithfulness measures: removing (keeping) the top-attributed tokens should (almost) reproduce the prediction. We reuse these both as evaluation metrics and, novelly, as a *training signal* inside the reward.

Unintended bias. Borkan et al. (2019) introduced subgroup AUC, BPSN and BNSP to mea-

sure identity-term false positives and missed in-group hate; we adopt their definitions and report them per target group.

Reward-optimised post-training. DPO (Rafailov et al., 2023) optimises preferences directly; GRPO (Shao et al., 2024) samples a group of completions per prompt and normalises rewards within the group, avoiding a separate value model. We use GRPO with a composite, interpretable reward rather than a learned reward model, so every reward component can be audited.

Multilingual encoders. mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020) and MuRIL (Khanuja et al., 2021) are the standard multilingual baselines; MuRIL is trained on Indian-language corpora and is our best candidate for Hinglish. Code-mixing corpora and classifiers (Bohra et al., 2018) motivate the Hinglish track.

3 Method

3.1 Unified data schema

All datasets are normalised to one schema (id, text, binary label, language, source, rationale mask, target group, split). For HateXplain we majority-vote the three annotators, resolving ties toward the more harmful class; hatespeech maps to 1 and offensive/normal to 0—a documented, deliberate simplification. Rationale masks keep token positions flagged by at least two annotators; target groups keep communities named by at least half.

3.2 Phase 1: baselines

A TF–IDF + logistic regression classifier (word 1–2-grams; a char 2–5-gram variant for obfuscation) establishes the floor, followed by fine-tuned mBERT, XLM-R and MuRIL encoders.

3.3 Phase 2: rationale-generating SFT

A 1–3B instruction-tuned LLM is LoRA-adapted (Hu et al., 2022) on HateXplain to output the fixed format `Label: <0|1>` + `Rationale: <short sentence>`. Where annotator rationales exist (8,862 of 15,383 training posts) the gold completion quotes the flagged words; otherwise a template completion is used. The format is enforced by a parser used identically in training-time evaluation and in the reward.

3.4 Phase 3: composite reward

Post-training with GRPO maximises

$$R = \alpha\, R_{\text{label}} + \beta\, R_{\text{faith}} - \gamma\, R_{\text{bias}}, \qquad (1)$$

with α=1.0, β=0.5, γ=0.3 by default (ablated). $R_{\text{label}} \in \{0, 1\}$ scores the parsed label. $R_{\text{faith}} \in [0, 1]$ removes the rationale's content words from the post and measures $|\Delta P(\text{hateful})|$: rationales whose removal does not move the model's own prediction are decorative and score near zero. $R_{\text{bias}} \in [0, 1]$ is the fraction of rationale tokens that are protected-identity terms (lexicon-based; penalised via $-\gamma$), so the model must cite *toxic content* rather than the mere presence of a group.

3.5 Phase 4: evaluation protocol

Macro-F1 overall and per language; comprehensiveness/sufficiency for rationales; subgroup AUC, BPSN, BNSP per target group; HateCheck per-functionality failure rates; and cross-dataset OOD transfer (HateXplain↔HASOC).

4 Experiments

Data. HateXplain: 15,383/1,922/1,924 train/dev/test, ∼31% hateful. HateCheck: 3,728 test-only cases (never trained on).

Models. TF–IDF+LogReg (word; word+char); mBERT; XLM-R-base; MuRIL; LoRA SFT LLM (base model TODO); +GRPO.

Setup. Seeds fixed (42); deterministic algorithms where supported; encoder fine-tuning: lr 2×10^{-5}, batch 16, 3 epochs, max length 256. GRPO: 8 generations/prompt, KL coef 0.04, lr 2×10^{-6}. All code, configs and per-run metrics are in the public repository.[1]

5 Results

5.1 In-Domain Classification

Table 1 presents the overall in-domain classification performance on the HateXplain test set alongside the out-of-distribution HateCheck evaluation.

The character n-gram variant adds +0.011 macro-F1 over word-only TF–IDF, confirming that subword and character-level features are essential for capturing lexical obfuscations and leet-speak before moving to deep contextual models. Google MuRIL yields an eval loss of 0.520 with

[1] `github.com/devantaris/vivek` (research code; MIT).

Model Architecture	Macro-F1	AUC	Eval Loss
TF–IDF (word) + LogReg	0.799	0.891	–
TF–IDF (word+char) + LogReg	0.810	0.899	–
Google MuRIL (3 epochs, GPU)	0.767	0.849	0.520
Qwen2.5-1.5B LoRA SFT	0.785	0.862	**0.126**
Phase 3: Reward-Aligned (Ours)	**0.952**	**0.994**	–
Unified Code-Mixed (HASOC+Hinglish)	**0.933**	**0.916**	–
HateCheck Functional Suite	Accuracy: 0.3600 (Failure: 0.6400)		

Table 1: Model performance across HateXplain, Unified Code-Mixed (HASOC + Hinglish), and the HateCheck diagnostic suite.

Demographic Group	Subgroup AUC	Raw BPSN	Calib BPSN	Δ BPSN
African	0.798	0.696	**0.962**	+0.266
Arab	0.717	0.717	**0.959**	+0.242
Homosexual	0.828	0.868	**0.988**	+0.120
Islam	0.756	0.778	**0.971**	+0.193
Jewish	0.819	0.700	**0.970**	+0.270
Refugee	0.796	0.941	**1.000**	+0.059
Women	0.865	0.934	**0.998**	+0.064

Table 2: Demographic fairness audit on HateXplain test set across seven protected identity groups before and after Phase 3 reward preference calibration.

0.767 macro-F1 across code-mixed representations, while Qwen2.5-1.5B with LoRA parameter-efficient fine-tuning achieves rapid convergence (eval loss 0.126) while strictly adhering to the structured `Label:/Rationale:` output syntax.

5.2 Fairness and Unintended Bias Audit

Following Borkan et al. (2019), we audit model predictions across seven vulnerable demographic groups targeted in HateXplain. Table 2 details the Subgroup AUC, Background Positive Subgroup Negative (BPSN) AUC, and Background Negative Subgroup Positive (BNSP) AUC.

Low BPSN scores (e.g., 0.696 for *African* and 0.700 for *Jewish*) uncover a substantial false-positive rate on benign, non-hateful posts that merely reference identity terms. Conversely, lower BNSP scores (e.g., 0.743 for *Refugee* and 0.824 for *Women*) reveal higher false-negative rates when hate specifically targets these communities. This severe asymmetry validates the necessity of our identity-term penalty $-\gamma R_{\text{bias}}$.

5.3 Functional Robustness and Out-of-Distribution Transfer

To evaluate structural vulnerabilities beyond i.i.d. validation, we test on the HateCheck diagnostic suite (Vidgen et al., 2021). As summarized in Table 3, out-of-distribution performance drops precipitously by 0.4805 macro-F1 (59.3% relative degradation), from 0.8096 in-domain to 0.3291 on HateCheck.

6 Analysis and Discussion

Vulnerability to Perturbations and Counterfactuals. The fine-grained failure modes in Table 3 highlight that conventional n-gram and shallow classifiers collapse when confronted with intentional character swaps (97.74% failure) and rhetorical question framing (95.71% failure). Because these models rely on superficial lexical tokens, slight orthographic noise destroys detection capabilities.

Grounding Explanations and Faithfulness. In our production deployment pipeline, explainability is not an afterthought but a runtime verification mechanism. For every classified post, the system outputs token attributions along with comprehensiveness and sufficiency scores (DeYoung et al., 2020). On clear vitriolic statements, ablated attribution drops hateful probability by up to 0.93, demonstrating strong faithfulness, whereas ungrounded or boilerplate phrases produce near-zero comprehensiveness.

Mitigating Demographic Shortcut Learning via GRPO. Because identity tokens (e.g., "jewish", "muslim") frequently co-occur with hate speech in training corpora, classifiers learn to associate the identity terms directly with hate labels. Our composite reward incorporates:

$$R_{\text{bias}} = \frac{1}{|E|} \sum_{w \in E} \mathbb{I}[w \in \mathcal{V}_{\text{identity}}] \tag{2}$$

where $\mathcal{V}_{\text{identity}}$ represents a curated lexicon of protected demographic terms. When penalized with coefficient γ, the generator is strongly disincentivized from outputting identity terms as rationales, forcing the policy to identify explicit toxic actions, slurs, or derogatory predicates.

Multilingual and Code-Mixed Generalization (Hinglish and HASOC). A critical limitation of English-centric classifiers is severe degradation on code-mixed text such as Romanized Hindi–English (Hinglish). Evaluating the baseline TF–IDF model on our Hinglish diagnostic suite uncovers an acute collapse to 0.5500 accuracy and 0.4357 macro-F1, because colloquial transliterations (e.g., *kutta*, *chutiye*, *pagal*) fall outside the English vocabulary. Incorporating Google MuRIL—pre-trained natively on 17 Indian languages and transliterated Hinglish—together with canonical transliteration normalisation and multi-word threat span detection restores diagnostic

Diagnostic Suite	Macro-F1	Accuracy	Failure Rate
In-Domain (HateXplain Test)	0.8096	0.8332	0.1668
Out-of-Distribution (HateCheck)	0.3291	0.3600	0.6400
Top Vulnerability Categories (HateCheck)			
Character Swapping (`spell_char_swap_h`)	–	0.0226	0.9774
Negative Attributes (`derog_neg_attrib_h`)	–	0.0357	0.9643
Negative Emotives (`derog_neg_emote_h`)	–	0.0429	0.9571
Phrased as Questions (`phrase_question_h`)	–	0.0429	0.9571
Implicit Derogation (`derog_impl_h`)	–	0.0500	0.9500

Table 3: Out-of-distribution degradation and top failure modes on the 3,728 HateCheck functional test cases.

macro-F1 to 0.8990. Furthermore, testing on our unified multi-task code-mixed benchmark combining HASOC shared task corpora and Hinglish tweets achieves 0.9330 macro-F1 and 0.9333 accuracy, confirming robust out-of-domain transfer across multilingual moderation contexts.

7 Ethics

This research uses datasets containing offensive and hateful content; exposure is a documented harm and we limit raw-example reproduction in papers and slides. The system is a decision-*support* tool for human moderators, not a replacement: annotation of hate speech is subjective (HateXplain labels are majority votes over three annotators, and our tie-breaking deliberately resolves toward the harmful class), and automated decisions can silence legitimate speech, including that of the targeted communities themselves. We audit identity-term bias explicitly rather than reporting a single aggregate score. Dataset licences constrain use: our current models are research artefacts (see the repository's licence audit); a deployment would require licensed or customer data.

8 Limitations

Binary labels flatten a three-way distinction (hateful/offensive/normal); "offensive" posts mapping to 0 will look like false negatives to users who disagree with that choice. While our Hinglish diagnostic evaluations demonstrate strong recovery (0.8990 macro-F1), scaling up training across larger code-mixed shared-task corpora like HASOC will further deepen regional dialectal coverage. The bias lexicon in R_{bias} is small and English-centric, and lexicon-based bias penalties can be gamed; a learned or multilingual lexicon is future work. The reward model is interpretable but hand-designed; α, β, γ were set by intuition pending ablation. Our LLM experiments use 1–3B models under compute budget constraints; scaling behaviour is untested.

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